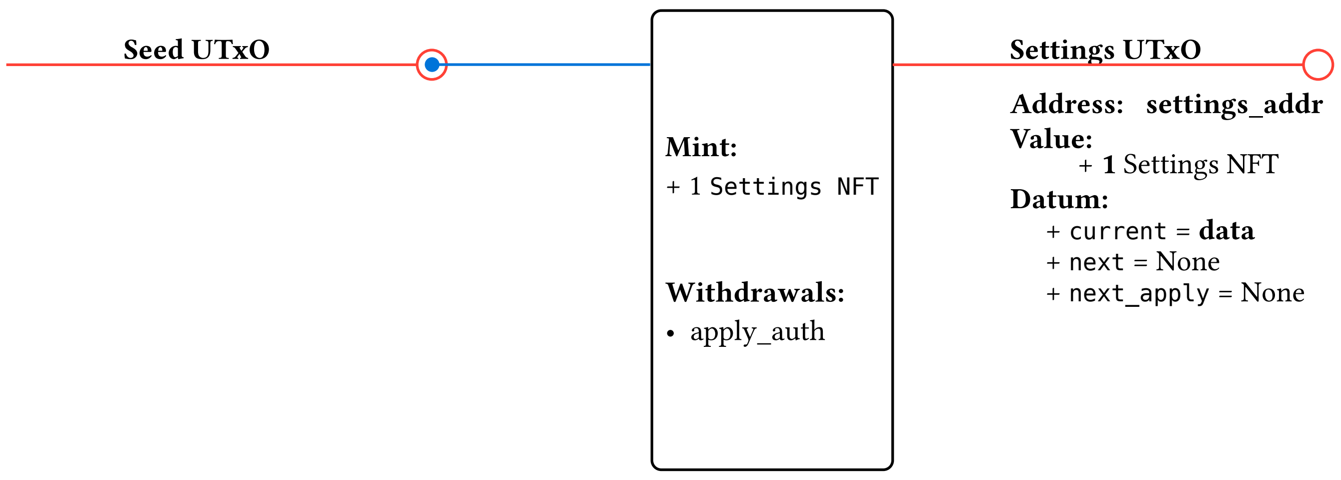
Contents

1 - Launch settings 2
2 - Propose settings 2
3 - Apply settings 2
4 - Close settings 2

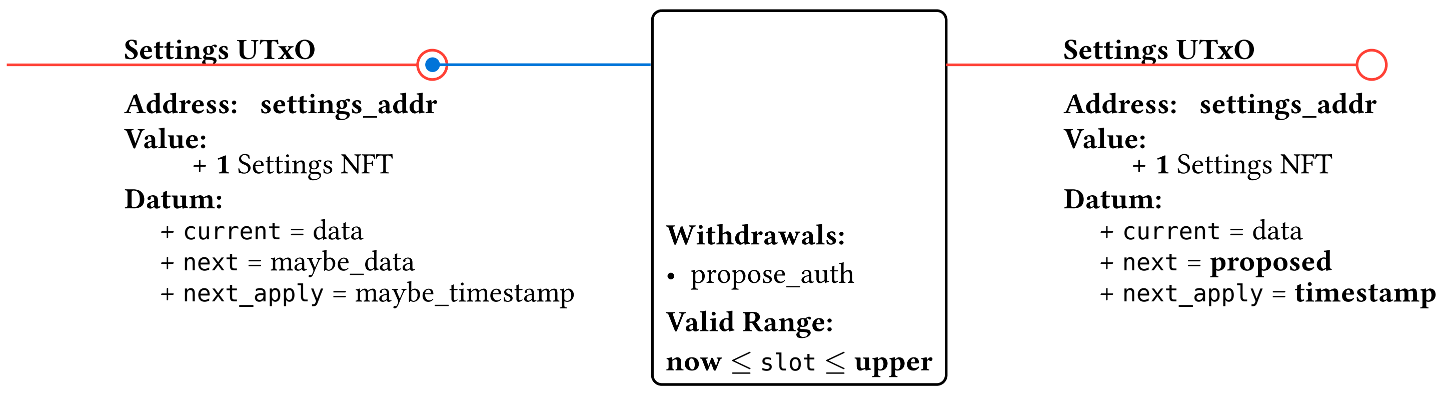
1 - Launch settings



Notes:
- `settings_addr`: payment is `settings_hash`, staking is any (chosen at launch)
- continuations must reproduce the spent input’s full address, so the staking key cannot be changed once deployed
- `settings_nft`: policy is `settings_hash`, token name is `settings_token_name`
- `settings_hash = hash(settings_validator(seed_utxo, propose_auth, apply_auth, apply_delay, settings_token_name))`
- `data`: the settings data. it is such that `validate_datum(data)` is true

Figure 1: Launch settings transaction

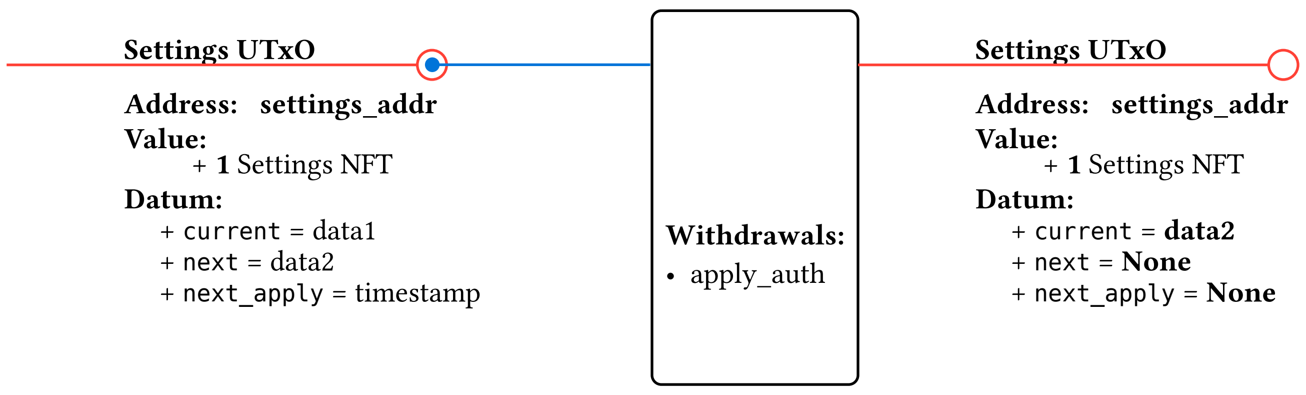
2 - Propose settings



Notes:
- `settings_addr`: payment is `settings_hash`, staking is unchanged
- `settings_nft`: policy is `settings_hash`, token name is `settings_token_name`
- `settings_hash = hash(settings_validator(seed_utxo, propose_auth, apply_auth, apply_delay, settings_token_name))`
- `maybe_data`: can be `None` or `Some(data)`
- `proposed`: the proposed settings data. it is such that `validate_datum(proposed)` is true
- `timestamp ≥ upper + apply_delay`, where `upper` is the propose transaction’s validity upper bound and `apply_delay` is the minimum time that must pass before the proposed settings can be applied. The ledger only includes the transaction within its validity range, so `upper` covers the inclusion slot (the reference builders set a bounded validity window and `timestamp = upper + apply_delay`)

Figure 2: Propose settings transaction

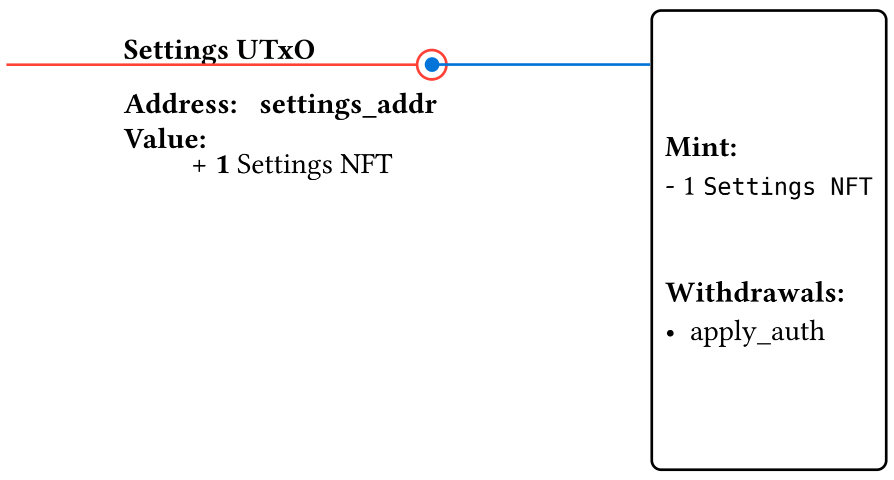
3 - Apply settings



Notes:
- `settings_addr`: payment is `settings_hash`, staking is unchanged
- `settings_nft`: policy is `settings_hash`, token name is `settings_token_name`
- `settings_hash = hash(settings_validator(seed_utxo, propose_auth, apply_auth, apply_delay, settings_token_name))`
- delay must have expired: `now >= timestamp`

Figure 3: Apply settings transaction

4 - Close settings



Notes:
- `settings_addr`: payment is `settings_hash`, staking is any
- `settings_nft`: policy is `settings_hash`, token name is `settings_token_name`
- `settings_hash = hash(settings_validator(seed_utxo, propose_auth, apply_auth, apply_delay, settings_token_name))`
- the `Settings NFT` must be burned (`quantity < 0`)
- the datum is not checked

Figure 4: Close settings transaction